Limits of Sequences: Solutions

Part A: L'Hôpital's Rule

1. $a_n = \dfrac{\ln n}{\sqrt{n}}$. The limit has the form ∞/∞, so with $f(x) = \dfrac{\ln x}{\sqrt{x}}$:

$$\lim_{x\to+\infty} \frac{\ln x}{\sqrt{x}} = \lim_{x\to+\infty} \frac{1/x}{\frac{1}{2}x^{-1/2}} = \lim_{x\to+\infty} \frac{2}{\sqrt{x}} = 0.$$

The sequence **converges to 0**.

2. $a_n = \dfrac{n^2}{e^n}$. Form ∞/∞; apply L'Hôpital's Rule twice:

$$\lim_{x\to+\infty} \frac{x^2}{e^x} = \lim_{x\to+\infty} \frac{2x}{e^x} = \lim_{x\to+\infty} \frac{2}{e^x} = 0.$$

The sequence **converges to 0**.

3. $a_n = \left(1 + \dfrac{3}{n}\right)^n$. This has the indeterminate form 1^∞. Let $y = \left(1 + \frac{3}{x}\right)^x$, so

$$\ln y = x \ln\left(1 + \frac{3}{x}\right) = \frac{\ln\left(1 + \frac{3}{x}\right)}{1/x},$$

which has the form $0/0$. By L'Hôpital's Rule,

$$\lim_{x\to+\infty} \ln y = \lim_{x\to+\infty} \frac{\dfrac{1}{1+3/x} \cdot \left(-\dfrac{3}{x^2}\right)}{-\dfrac{1}{x^2}} = \lim_{x\to+\infty} \frac{3}{1+3/x} = 3.$$

Hence $\lim_{x\to+\infty} y = e^3$, and the sequence **converges to** e^3.

Part B: The Squeezing Theorem

4. Numerically: $a_1 \approx 0.2919$, $a_{10} \approx 0.0704$, $a_{100} \approx 0.0072$, $a_{1000} \approx 0.0007$. Conjecture: the limit is 0.

Since $0 \le \cos^2 n \le 1$ for all n,

$$0 \le \frac{\cos^2 n}{n} \le \frac{1}{n}.$$

Both $0 \to 0$ and $\dfrac{1}{n} \to 0$, so by the Squeezing Theorem, $\displaystyle\lim_{n\to+\infty} \frac{\cos^2 n}{n} = 0$.

5. Numerically: $a_{10} \approx 0.4503$, $a_{100} \approx 0.4950$, $a_{1000} \approx 0.4998$. Conjecture: the limit is $\frac{1}{2}$.

Since $-1 \le \sin n \le 1$,

$$\frac{n-1}{2n+1} \le \frac{n+\sin n}{2n+1} \le \frac{n+1}{2n+1}.$$

Dividing numerator and denominator by n,

$$\lim_{n\to+\infty} \frac{n-1}{2n+1} = \lim_{n\to+\infty} \frac{1-1/n}{2+1/n} = \frac{1}{2}, \qquad \lim_{n\to+\infty} \frac{n+1}{2n+1} = \frac{1}{2}.$$

By the Squeezing Theorem, $\lim_{n\to+\infty} \dfrac{n+\sin n}{2n+1} = \dfrac{1}{2}$.

6. Numerically: $a_1 = 8$, $a_2 \approx 5.8310$, $a_5 \approx 5.0757$, $a_{10} \approx 5.0030$. Conjecture: the limit is 5.

Since $0 < 3^n \le 5^n$,

$$5^n \le 3^n + 5^n \le 2 \cdot 5^n.$$

Taking nth roots (which preserves inequalities between positive numbers),

$$5 \le (3^n + 5^n)^{1/n} \le 2^{1/n} \cdot 5.$$

Since $2^{1/n} \to 2^0 = 1$, the right side tends to 5, and the left side is the constant 5. By the Squeezing Theorem, $\lim_{n\to+\infty} (3^n + 5^n)^{1/n} = 5$.

Remark: the larger base dominates. In general, $(a^n + b^n)^{1/n} \to \max(a, b)$ for $a, b > 0$.

Part C: Limits of Recursively Defined Sequences

7.

a. Each term is $\sqrt{2 + (\text{previous term})}$, so

$$a_1 = \sqrt{2}, \qquad a_{n+1} = \sqrt{2 + a_n}.$$

b. Assume $a_n \to L$. Since $n + 1 \to +\infty$ as $n \to +\infty$, also $a_{n+1} \to L$. Taking the limit of both sides of the recursion,

$$L = \sqrt{2+L} \quad \Longrightarrow \quad L^2 = 2 + L \quad \Longrightarrow \quad L^2 - L - 2 = 0 \quad \Longrightarrow \quad (L-2)(L+1) = 0.$$

The solution $L = -1$ is extraneous because $a_n > 0$ for all n. Therefore $L = 2$.

8. Assume $a_n \to L$; then $a_{n+1} \to L$ as well. Taking limits,

$$L = \frac{1}{2}\left(L + \frac{5}{L}\right) \quad \Longrightarrow \quad 2L = L + \frac{5}{L} \quad \Longrightarrow \quad L = \frac{5}{L} \quad \Longrightarrow \quad L^2 = 5.$$

The negative solution is extraneous because $a_n > 0$ for all n (by induction: $a_1 = 1 > 0$, and if $a_n > 0$ then $a_{n+1} > 0$). Therefore $L = \sqrt{5}$.

Remark: this is the same "average a guess with 5 divided by the guess" idea as Example 10, here approximating $\sqrt{5} \approx 2.2361$: $a_2 = 3$, $a_3 \approx 2.3333$, $a_4 \approx 2.2381$, $a_5 \approx 2.2361$.

9. First five terms:

$$a_1 = 1, \quad a_2 = 2, \quad a_3 = \frac{3}{2}, \quad a_4 = \frac{5}{3}, \quad a_5 = \frac{8}{5}.$$

The numerators and denominators are consecutive Fibonacci numbers.

Assume $a_n \to L$; then $a_{n+1} \to L$. Taking limits,

$$L = 1 + \frac{1}{L} \quad \Longrightarrow \quad L^2 = L + 1 \quad \Longrightarrow \quad L^2 - L - 1 = 0 \quad \Longrightarrow \quad L = \frac{1 \pm \sqrt{5}}{2}.$$

The negative solution $\frac{1-\sqrt{5}}{2}$ is extraneous because $a_n \ge 1 > 0$ for all n. Therefore $L = \dfrac{1+\sqrt{5}}{2} \approx 1.618$, the golden ratio.